Conversion of Camphene to Camphor

Day 3: Conversion of Borneol to Camphor

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CHEM 2081-004

Introduction

In this experiment, the goal was to complete the third intermediate step of the overall reaction of camphene to camphor. That is, converting the borneol ($C_{10}H_{17}OH$) that was produced in the previous experiment to camphor ($C_{10}H_{17}O$) by reacting with Sodium Hypochlorite, or bleach (NaOCl) in order to reduce the hydroxide group to a ketone group on the carbon ring.

Materials and Methods

First, 0.294 g borneol was added to a flask along with 0.3 mL glacial acetic acid and 1 mL acetone. This solution was heated until all of the borneol was dissolved in the solution and the temperature reached 40-50 degrees celsius. After this, a total of 6 mL sodium hypochlorite (bleach) was added over the course of 35 minutes, while maintaining the 40-50 degree temperature. Next, the solution was allowed to cool, and the camphor product was extracted from the solution using two portions of DCM. The organic portion was neutralized using sodium bicarbonate, then washed again, this time using sodium bisulfate. The DCM portion, containing the desired camphor, was dried using sodium sulfate to remove any microscopic water droplets, and then evaporated over a water bath. This allowed the solid white camphor crystals to be left behind to be measured and observed.

Results

Product: 0.177 g camphor

Calculations

Theoretical yield: 0.294g borneol x 1 mol borneol/ 155.54 g borneol x 1 mol camphor / 1 mol borneol x 154.46 g camphor / 1 mol camphor = 0.292 g camphor

0.177g camphor / 0.292 g camphor x 100 = 60.63% yield

Discussion

During this experiment, 0.177 g of camphor was produced. There are a couple potential sources of error that could impact the yield that was produced. The primary potential source for error in this experiment is the potential for impurities in the initial borneol reactant. Although the borneol was completely white, which is a good indicator of purity, it is possible that minor impurities in the reactant would be present and would be able to decrease the yield of desired product. One important note is that solubility of various substances that are used in this reaction. Since there are many steps involving adding both organic and aqueous substances, including washing, neutralizing, and dissolving, it is important to understand why certain reactants make good or poor solvents. The primary reason for dissolution is “like dissolves like,” meaning that a polar substance (i.e. the aqueous layer) is able to dissolve polar or ionic compounds, while the nonpolar substance (the organic layer) is able to dissolve organic, nonpolar substances, including the camphor that was produced during the reaction.

References

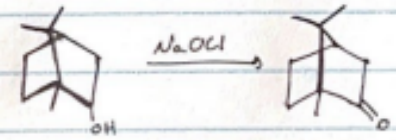
H McNeal. Conversion of Camphene to Camphor. *Chem 1081 Organic Chemistry II Lab Manual.* Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 69-70).

Camphene to Camphor : Borneol to camphor

1. in long neck 10 mL round bottom flask, add 0.3 g borneol w/ 0.3 mL glacial acetic acid. Dissolve in 1 mL Acetone. Add boiling chip
2. Assemble reflux, place apparatus in water bath. Maintain temp between 40-50°C, if borneol doesn't dissolve, add 0.5 mL acetone
3. Once reagents are dissolved, add 6 mL sodium hypochloride soln (bleach). Addition should be done through top opening of condenser. Add dropwise of 35 min. Swirl mixture gently + continue rxn 15 min after last bleach addition
4. Cool to room temp + extract camphor w/ 2 portions (2 mL each) of methylene chloride (or DCM). Use centrifuge to make more efficient.
5. Neutralize DCM layers w/ 2 mL of sodium bicarb. Once bubbling stops, soln is neutralized
6. Wash organic soln w/ 2 mL 5% sodium bisulfate + 2 mL water
7. Dry organic soln over anhydrous Na_2SO_4
8. Evaporate DCM over water bath. Weigh crystals + determine % yield

Observations

1 ✓ 0.294 g borneol



2 ✓

3 ✓

4 ✓ DCM @ bottom bc higher density

5 ✓

6 ✓ *pipette method for washing

7 ✓

8 ✓ watchglass: 33.150 g

watch glass + product: 33.327 g - white clear crystals

Lab Performance

Exp 6 D3

Alyssa Barrett | 2081-004 | 42

Name: | Section: | Room: | Hood:

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]